



AI-Assisted Queue Management

Predict. Route. Reduce Waiting.

An intelligent queue management system estimating real-time wait times and seamlessly routing visitors to the fastest counter.

Core Deck Snapshot

TARGET SECTORS

Hospitals • College Offices • Banks • Service Centers

CORE VALUE PROPOSITION

Eliminating visitor uncertainty & optimizing staff workloads

Queues Are an Information Problem

"Long wait times stem from zero real-time queue visibility — not just high footfall."



Zero Counter Visibility

Visitors cannot identify faster counters upon arrival.



Static Token Limits

Manual ticket systems offer no dynamic predictions.



Unbalanced Load

Staff overcapacity while adjacent counters stay idle.

THE CORE OBJECTIVE

Deliver clear visitor recommendations & real-time operational views without costly hardware.

Software-First AI Layer

02. OUR SOLUTION

Lightweight AI Queue Intelligence

Real-time queue tracking and predictive routing in 5 steps



Dynamic Multi-Counter Balancing

Continuously re-evaluates queue speeds and updates estimated completion times instantly.

ACTIVE ENGINE

Modular & Extensible Stack

User App / Web

View queue state, join token, and receive instant routing.

API / Backend

Authentication, queue status management, and business logic.

AI Analytics

Wait-time prediction model and dynamic counter routing algorithm.

Data Layer

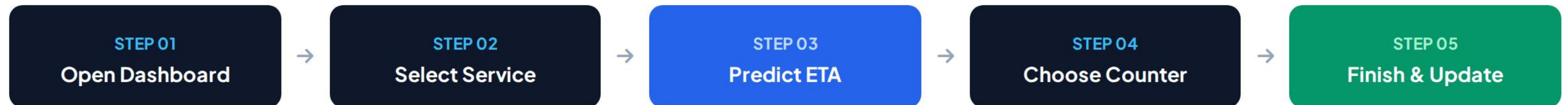
Queue timestamps, service event logs, and historical trends.



Privacy by Design

Stores only necessary queue event timestamps. No personally identifiable information (PII) required for prediction workflow.

Prototype & Demo Journey



🎬 Demo Walkthrough

Simulate a crowded counter surge → watch real-time ETA update dynamically → system re-routes new visitors to an available adjacent counter instantly.

🐙 GitHub Repository Deliverables

README Setup

Source Code

Sample Queue Data

Dashboard Screenshots

Feasibility & Risk Mitigation

IDENTIFIED RISK	ACTIONABLE MITIGATION
Inaccurate early predictions due to sparse historical data	Deploy baseline rule engine + average service-time baseline; refine models dynamically as events accumulate.
Peak traffic causes stale queue state latency	Enforce timestamped event streams and high-frequency live dashboard refresh.
User skepticism regarding predicted ETA accuracy	Display transparent confidence ranges & expose live queue state context alongside raw ETA.
Escalating deployment & hardware overheads	Maintain software-first MVP; integrate IoT/vision sensors selectively where ROI is proven.

Scaling SmartQueue AI

Measurable Impact

Reduced Wait Times: Lowers perceived and actual queue delays.

Visitor Distribution: Prevents single-counter crowding.

Staff Visibility: Live load telemetry for floor management.

Multi-Sector Deploy: Configurable for clinics, banks, offices.

Next Milestones

ML Optimization: Train prediction on live event logs.

Push Alerts: Instant mobile token queue notifications.

Role Dashboards: Specialized manager vs clerk views.

Pilot Testing: Measure wait reduction in active sites.

SUBMISSION REPOSITORY

<https://github.com/patilavi127-stack/medsafeAi>

Build small. Measure real queues. Improve every iteration.

Image Sources



<https://qwaiting.com/website/asset/images/home/home-banner-image.webp>

Source: qwaiting.com